Yuvaraj A

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PROFESSIONAL SUMMARY

Detail-oriented and motivated **AI Developer/Data Scientist** graduate from the **TNSDC-Cultus AI program**. Possess a strong academic foundation in Data Science, Machine Learning, Deep Learning, Python. Hands-on experience through academic and personal projects involving Data cleaning, Exploratory Data Analysis (EDA), feature engineering and Model Evaluation using Pandas, NumPy, Scikit-learn, TensorFlow. Familiar with data visualization and basic NLP concepts. Actively seeking an Entry-level Data Scientist, AI/ML or Python Developer role to solve real-world business problems using data-driven approaches.

TECHNICAL & SOFT SKILLS

- **Programming Languages**: Python, SQL.
- **Web Technologies:** HTML, CSS, JavaScript.
- **AI / Machine Learning & Data Science**: TensorFlow, Keras, Scikit-Learn, Pandas, NLP Basics.
- **AI & ML Algorithms**: Regression, Classification, Clustering, Neural Networks.
- **Tools & Platforms**: Git, GitHub, Jupyter Notebook, VS Code, Google Colab, Excel, Power BI.
- **Soft skills**: Problem solving, Team Collaboration, Time management, Adaptability.
- **Deployment**: Streamlit, Flask.

PROJECTS

1. Title: FactoryGuard AI – Predictive Maintenance System | **GitHub (Source Code)** | **Live Demo**

Tools & Libraries used : Python, Matplotlib, NumPy, Pandas, Scikit-Learn.

Responsibilities :

- **Situation**: Machine breakdowns occurred without early warning, leading to unplanned downtime.
- **Task**: Develop a machine learning model to predict failures within 24 hours with high precision to minimize false alarms.
- **Action**: Engineered time-series features (rolling stats, EMA, lag), tuned XGBoost using Optuna, optimized precision-based threshold, implemented SHAP explainability, and deployed via Flask.
- **Result**: Achieved ≥90% precision on test data, enabling reliable early failure detection and reduced false alerts.

2.Title : Smart Agriculture – Crop Yield Prediction | **GitHub (Source Code)** | **Live Demo**

Tools & Libraries used : Python, TensorFlow/Keras, Matplotlib, NumPy, Pandas, Streamlit, Scikit-Learn.

Responsibilities **:**

- **Situation**: Crop yield prediction was inaccurate due to complex soil and crop data patterns.
- **Task**: Build an ANN regression model to predict crop yield accurately and make it usable in real time.
- **Action**: Performed EDA, trained an ANN model using TensorFlow/Keras, and deployed it via Streamlit.
- **Result**: Achieved a high R^2 Score with low MAE, delivering a real-time crop yield prediction system.

EDUCATION

Master of Science, Physics | 2023-2025 | CGPA – 7.46 |
N.M.S.S. Vellaichamy Nadar College, Madurai, Tamil Nadu.

Bachelor of Science, Physics | 2020-2023| CGPA – 7.54 |
N.M.S.S. Vellaichamy Nadar College, Madurai, Tamil Nadu.

CERTIFICATIONS

- **TNSDC AI Certification** , Cultus & TNSDC - View Certificate
- **AWS Educate Machine Learning Foundations**, Amazon Web Services (AWS) - View Credential
- **AWS Educate Introduction to GenAI**, Amazon Web Services (AWS) - View Credential
- **Data Analytics**, Novitech Private Limited - View Certificate
- **Artificial Intelligence**, Novitech Private Limited - View Certificate